KOPPARTHI LAKSHMI SUKADHA
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Data Science

Professional Summary

Data Science and AI enthusiast with strong expertise in Python, Machine Learning, Deep Learning, NLP, and Generative AI, currently pursuing B.Tech in Computer Science. Experienced in developing predictive models, AI agents, deep learning classification systems, and Flask-based AI applications. Hands-on experience with RAG, LLMs and data visualization tools, delivering high-accuracy models for real-world datasets, including medical imaging and sentiment analysis. Passionate about building scalable, data-driven solutions using advanced ML and AI techniques.

Education

- **SRM Institute of Science and Technology**
  Master of Business Administration in Artificial Intelligence and Data Science
  CGPA: 9.4/10
- **Sagi Rama Krishnam Raju Engineering College**
  Bachelor of Technology in Computer Science and Engineering 2025 | CGPA: 9.22/10
- **Star Junior College**
  Intermediate (MPC)
  2021 | CGPA: 8.5/10
- **Nirmala High School**
  Board of Secondary Education
  2019 | CGPA: 8.5/10

Core Competencies

- **Programming & Development:** SQL, Python, Machine Learning, EDA, Deep Learning, NLP, RAG, LLMs, Generative AI
- **Machine Learning & Deep Learning:** Logistic Regression, Linear Regression, Decision Trees, Random Forest, SVM, KNN
- **Data Analysis Libraries:** NumPy, Pandas, Matplotlib, Seaborn
- **Soft Skills:** Problem-solving, Leadership, Attention to detail, Collaboration, Innovative, Teamwork, Communication, Time management

Internship Experience

LLMs and AI Agents Intern | Yaanthra Company | Apr 2024 – Oct 2024 | Hyderabad

- Developed 3+ AI agents to automate task-specific workflows, improving response efficiency by approximately 30%.
- Applied Retrieval Augmented Generation (RAG) techniques to improve response accuracy by 25% and reduce irrelevant outputs by 18%, enhancing contextual information retrieval.
- Enhanced Large Language Models (LLMs) to improve response coherence and reduce bias in generated outputs.

AI-ML Intern | Henotic Technology Pvt Ltd | Jan 2024 – Mar 2024 | Hyderabad

- Built and assessed 5+ machine learning models using Python, improving prediction accuracy by up to 18% through feature extraction and model tuning.
- Implemented Logistic Regression, Decision Trees, and Random Forest on structured datasets, increasing classification performance compared to baseline models.
- Performed data preprocessing and analysis on large datasets, reducing data inconsistencies by 25% through cleaning and normalization techniques.
- Delivered predictive analytics insights that improved model interpretability and evaluation efficiency.

Projects

Employee Promotions Prediction using Machine Learning
Technologies: Python, Oracle, Jupyter Notebook, Logistic Regression, Decision Trees, LightGBM

- Designed a predictive model to determine employee promotion eligibility using structured HR data of 5,000+ employee records, achieving 90% classification accuracy.
- Applied and compared Logistic Regression, Decision Trees, and LightGBM models, improving prediction performance by 15% over baseline models.
- Performed data preprocessing, feature extraction, and exploratory data analysis using Python and Oracle, enhancing model efficiency by 20%.
- Assessed models using accuracy, precision, recall, and F1-score, optimizing hyperparameters to reduce misclassification rate by 12%.

Orthopedic Image Classification – Developmental Dysplasia of Hip (DDH)
Technologies: Python, Image Processing, Machine Learning

- Processed and analyzed 1,000+ hip X-ray images, achieving 88% classification accuracy in distinguishing Normal and DDH cases.
- Applied image preprocessing techniques such as normalization and feature extraction, increasing classification precision by 14%.
- Applied machine learning algorithms to enhance diagnostic prediction reliability and reduce false positives by 12%.
- Analyzed model performance using accuracy and validation metrics, optimizing parameters to improve overall model stability and consistency.

Sentiment Analysis on Smartphone Reviews
Technologies: Python, BiLSTM, CNN, NLP

- Designed deep learning models for sentiment classification on 10,000+ smartphone reviews, achieving 92% accuracy using BiLSTM architecture.
- Executed NLP preprocessing techniques including tokenization, stopword removal, and text normalization, improving model performance by 12%.
- Compared CNN and BiLSTM models using precision, recall, and F1-score, selecting the optimal model, which improved overall classification performance by 8% over the baseline.
- Optimized hyperparameters and applied Dropout and MaxPooling layers, reducing overfitting by 15% and improving generalization capability across validation data.

AI-Powered Spam Email Detection Chatbot
Technologies: Python, NLP, TF-IDF, FAISS, RAG, Sentence Transformers, Gemini 2.5 Flash

- Developed an AI-powered spam detection chatbot using NLP, TF-IDF, and LinearSVC to classify email/message content as SPAM or NOT SPAM through a Flask REST API.
- Implemented an end-to-end RAG pipeline using Sentence Transformers and FAISS to retrieve relevant cybersecurity knowledge on spam indicators, phishing, and safe email practices.
- Integrated Gemini 2.5 Flash to generate contextual explanations and safety recommendations by combining the ML prediction with retrieved RAG knowledge, improving the interpretability of spam detection results.
- Built an interactive React.js chatbot interface integrated with Flask, displaying the prediction, model score, AI-generated explanation, and retrieved knowledge in real time.

Achievements & Certifications

- NPTEL Python for Data Science (Elite)
- AWS Machine Learning Foundations
- SQL (Basic) Certificate- HackerRank
- Generative AI Fundamentals - LinkedIn Learning
- **Hackathons:** Prajwalan 2k23 (College), HACKOVERFLOW-2k23 (SRKR Coding Club), Prajwalan 2k24
- **Workshops:** Android Development, Computer Vision,Mobile App Development,GEN AI Fundamentals Course (participation Certificate).